**PH.30102** **FALL 2026**

# Quantum Mechanics III

## Course Syllabus

Seonwoo Park Fall 2026

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|---|---|
| Instructor | Seonwoo Park |
| Teaching assistants | Arin Jeon, Doha Nam |
| Semester | Fall 2026 |
| Language | English |

### Course description

Advanced quantum mechanics is developed through quantum states, channels, generalized measurements, open-system dynamics, scattering, tomography, and parameter estimation.

### Learning outcomes

- Model preparations and measurements with density operators and POVMs.
- Analyze quantum channels with Kraus and Choi representations.
- Solve Lindblad and stochastic master equations.
- Reconstruct states and processes with statistically valid uncertainty estimates.

### Assessment

| Component | Weight |
|---|---|
| Problem sets | 35% |
| Tomography laboratory | 15% |
| Midterm examination | 20% |
| Final project | 20% |
| Participation | 10% |

### References


- M. Nielsen and I. Chuang, Quantum Computation and Quantum Information, selected operational chapters.
- H. M. Wiseman, Quantum Measurement and Control, campus selections.

- S. Park, Quantum Mechanics III: Preparations, Channels, and Measurements, lecture notes rev. 12.
- National Instrument Registry, Quantum Device Calibration Table QDC-26.

## Weekly schedule

| Week | Dates | Topic |
|---|---|---|
| 1 | 2026-08-31–2026-09-06 | Quantum States and Preparation Procedures |
| 2 | 2026-09-07–2026-09-13 | Generalized Measurements and Instruments |
| 3 | 2026-09-14–2026-09-20 | Quantum Channels and Kraus Representations |
| 4 | 2026-09-21–2026-09-27 | Open Systems and Lindblad Generators |
| 5 | 2026-09-28–2026-10-04 | Scattering and Measured Cross Sections |
| 6 | 2026-10-05–2026-10-11 | Identical Particles and Count Correlations |
| 7 | 2026-10-12–2026-10-18 | Quantum State Tomography |
| 8 | 2026-10-19–2026-10-25 | Midterm Review |
| 9 | 2026-10-26–2026-11-01 | Correlation Experiments |
| 10 | 2026-11-02–2026-11-08 | Continuous Measurement and Quantum Trajectories |
| 11 | 2026-11-09–2026-11-15 | Parameter Estimation and Experimental Design |
| 12 | 2026-11-16–2026-11-22 | Control and Error Channels |
| 13 | 2026-11-23–2026-11-29 | Quantum Process Tomography |
| 14 | 2026-11-30–2026-12-06 | Automated Instrument Calibration |
| 15 | 2026-12-07–2026-12-13 | Blind-State Characterization |

## Submission policy

Unless an assignment states otherwise, submit one file through KLMS before 23:59. Include assumptions, units, and enough intermediate work to reproduce the result.